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CMPT 365 Written Assignment 1

1.5 Q2

Three novel applications of multimedia are education, medical, and virtual reality. In the field of education, multimedia has made education more accessible and interactive. Multimedia has made it easier for students to learn from home. Additionally, many students are learning better as a result of having videos and images to better explain the material. In the medical field, multimedia and the faster internet are allowing doctors to give orders to patients even from a distance. This could help those living in rural areas where there are not that many doctors. Advanced computer graphics could also help with diagnosing diseases. Lastly, multimedia is creating new styles of entertainment with virtual reality. Virtual reality is creating new experiences which make users feel more connected with multimedia such as video and audio.

6.4

3. $20 = 10\log x$ $\quad \log x = 2 \quad x = 100$

8. $20\log\left(\frac{V_{max}}{V_{min}}\right) = 60 \quad \frac{V_{max}}{V_{min}} = 10^3 = 1000 \quad 2^{10} = 1024$ and $2^{16} = 65{,}536$

Both are greater than 1000 so there is enough ~~to~~ for the signal.

12. a) $\frac{1}{2 \cdot 0.001} = \frac{1}{0.002} \Rightarrow 500\,Hz$ b) $\frac{1}{2 \cdot 0.005} = \frac{1}{0.01} \Rightarrow 100\,Hz$ c) 3600 sec in 1h,

$\frac{1}{2 \cdot 3600} = \frac{1}{7200} \Rightarrow 1.389 \times 10^{-4}\,Hz$

4.4

8. a) If an image is not correctly gamma-corrected, the image will appear dark. b) Gamma correction will increase brightness. If it is increased too much, the image will become too white. The levels for each colour will be raised to increasing γ. The greater the γ, the brighter the image.

9. a) This reduces the number of available colours. Some of the original values that are similar will be mapped to the same value.

b) The low end near 0 is more affected by gamma correction. This is because $\sim\frac{1}{2}$ is the same as a square root and square roots affect numbers that are smaller more. $R=1$ gets mapped to 16 while $R=254$ gets mapped to 254.

5.7

11. 4:2:2 : 2 samples for every 4 luma samples $\quad 4K: 3840 \times 2160 \times 12 \times 120 \times 2$
$= 2.389 \times 10^{10}$ bps $\quad 8K: 7680 \times 4320 \times 12 \times 120 \times 2 = 9.555 \times 10^{10}$ bps
$16K: 15360 \times 8640 \times 12 \times 120 \times 2 = 3.822 \times 10^{11}$ bps

14. b) $MSE = \frac{8(60-50)^2 + 8(80-50)^2 + 8(170-200)^2 + 8(190-200)^2}{64} = 250$

$PSNR = 10\log_{10}\frac{255^2}{250} \approx 24.15$ c) $MSE\left[8(60-50)^2 + 8(60-50)^2 + 8(70-50)^2 + 8(90-50)^2 + 8(180-200)^2 + 8(200-200)^2 + 8(200-200)^2 + 8(210-200)^2 + 8(210-200)^2\right]/64$
$= 350 \quad PSNR = 10\log_{10}\frac{255^2}{350} \approx 22.6901$

EX1. a) $65 + 10\log 2 \approx 68\,dB$ b) $65 + 10\log 5 \approx 72\,dB$ c) $65 + 10\log 200 \approx 88\,dB$

EX2. a) $0 = -10 + 10\log x \quad 10 = 10\log x \quad 1 = \log x \quad x = 10$ b) $-10 + 10\log 300 \approx 14.8$ dB

EX3 a) $20\log\frac{3}{V_{thresh}} = 65 \quad \log 3 - \log V_{thresh} = 3.25 \quad \log V_{thresh} = \log 3 - 3.25$

$V = \frac{3}{10^{3.25}} \approx 0.00169 \quad 20\log v - 20\log(0.00169) = 71 \Rightarrow 20\log v \approx 15.5577$

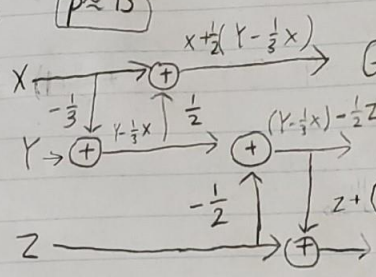
b) Sound of 4 violins: $65 + 10\log 4 \approx 71 \quad \log v = 0.77789 \quad \boxed{v \approx 6.0\,mV}$

EX4 When $U=V=0$, the pixel is grey, so $R=G=B=134$

EX5 $67 = 255\left(\frac{P}{255}\right)^{\frac{1}{2.2}} \Rightarrow 67^{2.2} = 255^{2.2}\,\frac{P}{255}$ $\quad 10408 \approx 255^{1.2}P$

$P \approx \frac{10408}{255^{1.2}}$ $\quad \boxed{P \approx 13}$

EX6.



$G = X + \frac{1}{2}(Y - \frac{1}{3}X) = \frac{6}{6}X + \frac{1}{2}Y - \frac{1}{6}X = \frac{5}{6}X + \frac{1}{2}Y$

$B = (Y - \frac{1}{3}X) - \frac{1}{2}Z = -\frac{1}{3}X + Y - \frac{1}{2}Z$

$R = Z - \frac{1}{3}X + Y - \frac{1}{2}Z = -\frac{1}{3}X + Y + \frac{1}{2}Z$

$$\begin{bmatrix} \frac{5}{6} & \frac{1}{2} & 0 \\ -\frac{1}{3} & 1 & -\frac{1}{2} \\ -\frac{1}{3} & 1 & \frac{1}{2} \end{bmatrix}$$